3.10.6 Radionuclide imaging and therapy (A-level only)

3.10.6.1 Imaging techniques (A-level only)

Content
Use of a gamma-emitting radioisotope as a tracer; technetium-99_m, iodine-131 and indium-111 and their relevant properties. The properties should include the radiation emitted, the half-life, the energy of the gamma radiation, the ability for it to be labelled with a compound with an affinity for a particular organ. The Molybdenum-Technetium generator, its basic use and importance. PET scans.

3.10.6.2 Half-life (A-level only)

Content
Physical, biological and effective half-lives; $\frac{1}{T_E} = \frac{1}{T_B} + \frac{1}{T_P}$; definitions of each term.

Basic structure and workings of a photomultiplier tube and gamma camera.

3.10.6.4 Use of high-energy X-rays (A-level only)

Content
External treatment using high-energy X-rays. Methods used to limit exposure to healthy cells.

3.10.6.5 Use of radioactive implants (A-level only)

Content
Internal treatment using beta emitting implants.

3.10.6.6 Imaging comparisons (A-level only)

Content
Students will be required to make comparisons between imaging techniques. Questions will be limited to consideration of image resolution, convenience and safety issues.